

**SECTION A**

General Unit Information:

Unit Name	Web Technologies And Platforms - COPY
Unit Code	CIS051-2
Level	5
Credit Value	30
Location(s) of Delivery	On Campus
Period(s) of Delivery	11 weeks
Pre-requisites or Restrictions	N/A
Aims and Relevance	You will learn how to use client side and server side technologies to implement web applications. Issues of design and usability will be explored and you will produce and validate your own product designs prior to implementing them using current development tools. The primary aims of the unit are: 1. To equip students with the technical understanding needed for the design and development of web applications 2. To develop practical skills in the design, implementation and evaluation of web based applications.
Syllabus Content	Overview of web technologies and standards: • Web formats, scripting and mark-up languages - such as: HTML5, JavaScript, Typescript, CSS3, XML • W3C standards, HTTP protocol, the Document Object Model (DOM) • Web servers and database integration • Open source web server standards - LAMP (Linux Apache, MySQL, PHP) • Data management and content management systems • Installation and maintenance of open-source CMS for promotion, learning or social applications – for example, WordPress • Design methods for web applications, usability, and testing • Responsive web design • Deploy a variety of web development tools and web technologies • Web application architectures • Model-View Controller Architecture • Dependency management – Composer, Bower, pip • The choice of modern web application frameworks frontend, backend and full stack – for example, Ruby and Ruby on Rails, ReactJS, Express, Angular, Codelgniter. • Implementing Design Patterns • Implement REST Web APIs and RESTful Web Services • ASP.NET Core technologies • Create a Web application with ASP.NET Core MVC and the MVC pattern • Create a Razor Pages Web application with ASP.NET Core

Learning Outcomes

On completion of this unit you should be able to

1. Demonstrate the following knowledge and understanding

Exploit and critique the principal technologies and concepts that are associated with developing web applications.

2. Demonstrate the following skills and abilities

Deploy complex skills for modifying and developing web applications using relevant server side and client side scripting and mark-up languages that can solve a range of real-world problems

PSRB Outcomes and/or Apprenticeship Standard

N/A

Summary Learning Hours

	Scheduled	Guided	Independent	Autonomous	Placement	Total
Hours	72	51	132	45	0	300
Percentage	24%	17%	44%	15%	0%	100%

Approach to Learning

This unit will be delivered using formal lectures, lab-based practical exercises, and tutorial sessions to help students to understand the processes of the digital creative industries and develop their problem-solving and professional skills. The following specific elements are included:

1. Lectures introduce the practices of the digital creative industries in relation to business environments and their operations, and the professional processes of digital creative technologies
2. Lab-based practical sessions and tutorials provide students with opportunities of consolidating the concepts and techniques introduced in the lectures with the aim of applying them to develop solutions to problems based on real-world scenarios.
3. Guided and self-directed review of published literature, e.g. books, research papers, web-based resources including tutorials on the tools and techniques, and products used for digital creative industry production. The use of electronic and paper based resources is expected and encouraged.

Assessment Summary

No	Assessment Method	Learning Outcome(s)	Weight	Submission Week	Length of Assessment
1	Coursework - Case Study	1 and 2	30	8	2000 words
2	Coursework - Artefact	1 and 2	70	13	na

Learner Development

This Unit particularly focuses on the development of your abilities in the following areas:

Enquiry	You will research and identify current design and technology trends in the creative industries. You will develop an interactive product based on a brief which will require you to research and experiment and apply and document the techniques used and knowledge you gain. You will develop experimental prototypes which you will test and further refine through an iterative development process.
Contextual Understanding, including where employer based	You will explore the range of application areas for digital media, and understand the different constraints that exist in each area. You will develop skills in content creation using industry relevant packages. You will develop understanding and apply the design, pre-production and usability techniques and processes used in the creative interactive technologies sector.
Collaboration	You will work as part of a team to develop a product to meet the needs of a given brief.
Enterprise	You will have the opportunity to creatively explore new ways to produce interactive assets and products in response to an open brief. You will take into account real-world constraints while developing innovative and creative solutions.
Assessment 1 Details	You will apply your knowledge of web technologies and techniques to propose and present a solution to a given commercial or technical problem.
Assessment 2 Details	You will submit an artefact (web-based software solution) as specified in the format outlined in the detailed assessment brief for the unit that makes use of knowledge and skills relating to web technologies.
Assessment 3 Details	
Assessment 4 Details	

Threshold Expectations

In order to pass Assessment 1 you will need to :

Outline a proposed solution in response to problem scenario given to you by the tutor

Justify the selection of a web technology or technique for addressing the problem scenario given to you by the tutor.

Explain the key concepts needed to develop a web application (best practice)

Provide a plan for the evaluation at different stages.

In order to pass Assessment 2 you will need to:

Identify a suitable technology to address a particular commercial or technical requirement

Develop client and server side solutions via hard coding and/or via the use of a software tool

Critically reflects on a solution that has been developed.

In order to pass Assessment 3 you will need to:

In order to pass Assessment 4 you will need to:

Section B

Recommended Reading

Core Text - this unit is supported by the following core text:

See <https://bedfordshire.rl.talis.com/units/cis051-2.html>

Guided Reading

The following is expected reading for this unit. Details of what to read and when will be provide in the BREO site.

See <https://bedfordshire.rl.talis.com/units/cis051-2.html>

Independent Study

To receive high grades you will need to demonstrate your wider reading. The following resources provide useful background reading for the material in this unit. This is not an exhaustive list and students should read widely from the variety of journals available in the Learning Resources Centre.

See <https://bedfordshire.rl.talis.com/units/cis051-2.html>

Equality Impact Assessment

Question	Y/N/NA	Additional anticipatory adjustments/actions if necessary
Learning materials will be made available in advance of sessions for students to adapt as appropriate?	Y	
The approach to teaching and learning is sufficiently flexible to enable all students to succeed?	Y	
The approach to group work takes account of the needs of students with disabilities and from diverse backgrounds?	Y	
The approach to practical work takes account of the needs of students with disabilities?	Y	
Students with a protected characteristic* have an equal opportunity to achieve the learning outcomes?	Y	
The assessment tasks provide all students with an equal opportunity to succeed?	Y	
Any other aspects of the unit that might pose potential challenges from an equality or diversity perspective have been considered?	Y	

* Age, Gender reassignment, Marriage and civil partnership, Pregnancy and maternity, Race, Religion and belief, Sex, Sexual orientation

Section C

Administrative Information – Faculty completion	
Faculty	Creative Arts Technologies and Science
Portfolio	CU-COM
School/Department	School of Computer Science and Technology
HECOS Code(s)	100375
Unit Co-ordinator	Susan Brandreth
Apprenticeship ?	No

Courses that this unit is delivered on
BSc (Hons) Computer Science
BSc (Hons) Software Engineering
BSc (Hons) Information Technology
BSc (Hons) Computer Games Technology